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ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60 (N&S), Elite, IAS & LIIT **JEE-MAIN-Shift-2** **Date: 19-03-2026**

Time: 03.00Pm to 06.00Pm **GTM-24** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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19-03-2026_Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main- Shift-2_GTM-24_Test Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let $f(x)=\begin{cases}4x-x^3+\ln\left(a^2-3a+3\right) & ,\ \text{if } 0\le x<3\\ x-18 & ,\ \text{if } x\ge 3\end{cases}$ complete set of value of 'a' such that f(x) has a local minimum at x=3 is

1) $[-1,2]$ **2)** $(-\infty,1)\cup(2,\infty)$ **3)** $[1,2]$ **4)** $(-\infty,-1)\cup(2,\infty)$

2. If the sum of all rational terms in the expansion of $\left(2^{\frac{1}{5}}+5^{\frac{1}{3}}\right)^{15}$ is K, then sum of all digits of K is

1) 10 **2)** 12 **3)** 13 **4)** 11

3. Domain of $f(x)=\sqrt{[x]-1+x^2}$ is ____(where [.] is GIF)

1) $(-\infty,-2)\cup[1,\infty)$ **2)** $(-\infty,-\sqrt{2})\cup[1,\infty)$ **3)** $(-\infty,-3)\cup[1,\infty)$ **4)** $(-\infty,-\sqrt{3}]\cup[1,\infty)$

4. The parabola $y^2=4x$ and the circle $(x-6)^2+y^2=r^2$ will have no common tangent if r is

1) $r>\sqrt{20}$ **2)** $r<\sqrt{20}$ **3)** $r>\sqrt{18}$ **4)** $r\in(\sqrt{20},\sqrt{28})$

5. The 20th term from the end of the progression $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots\ldots, -129\frac{1}{4}$ is

1) -115 **2)** -118 **3)** -110 **4)** -100

6. Let the image of the point $(1,0,7)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ be the point (α,β,γ). Then which one of the following points lies on the line passing through (α,β,γ) and making angles $\frac{2\pi}{3},\frac{3\pi}{4}$ with y-axis, z-axis respectively and an acute angle with x-axis is ___?

1) $(3,4,3-2\sqrt{2})$ **2)** $(3,-4,3+2\sqrt{2})$ **3)** $(1,-2,1+\sqrt{2})$ **4)** $(1,2,1-\sqrt{2})$

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7. The position vectors of the vertices A,B and C of a triangle are $2\hat{i}-3\hat{j}+3\hat{k}, 2\hat{i}+2\hat{j}+3\hat{k}$ and $-\hat{i}+\hat{j}+3\hat{k}$ respectively. Let ℓ denote the length of angle bisector AD of $\angle BAC$ where D lies on the line segment BC then $2\ell^2$ equals

1) 50 **2)** 42 **3)** 45 **4)** 49

8. If $2\tan^2\theta - 5\sec\theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$ for the least value of $n \in N$ then $\sum_{k=1}^{n} \frac{k}{2^k}$ is equal to

1) $\frac{1}{2^{13}}\left(2^{14}-15\right)$ **2)** $1-\frac{15}{2^{13}}$ **3)** $\frac{\left(2^{14}-14\right)}{2^{15}}$ **4)** $\frac{\left(2^{15}-15\right)}{2^{14}}$

9. Let $f:\left[\frac{-\pi}{2}, \frac{\pi}{2}\right] \to R$ be a differentiable function such that $f(0) = \frac{1}{2}$. If $\lim_{x\to 0} \frac{x\int_0^x f(t)dt}{e^{x^2}-1} = \alpha$ then $8\alpha^2$ is equal to

1) 16 **2)** 2 **3)** 1 **4)** 4

10. Consider the system of linear equations $x+y+z=4\mu$, $x+2y+2\lambda z=10\mu$, $x+3y+4\lambda^2 z=\mu^2+15$ where $\lambda, \mu \in R$, which of the following is not correct.

1) The system has unique solution if $\lambda \neq \frac{1}{2}, \mu \neq 15$

2) The system is inconsistent if $\lambda = \frac{1}{2}, \mu \neq 1$

3) The system has infinite number of solutions it $\lambda = \frac{1}{2}, \mu = 15$

4) The system is consistent if $\lambda \neq \frac{1}{2}$

11. Let M denote the median of the following frequency distribution

Class	0-4	4-8	8-12	12-16	16-20
Frequency	3	9	10	8	6

Then 20M is equal to

1) 416 **2)** 104 **3)** 52 **4)** 208

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12. If $f(x)=\begin{vmatrix} 2\cos^4 x & 2\sin^4 x & 3+\sin^2 2x \\ 3+2\cos^4 x & 2\sin^4 x & \sin^2 2x \\ 2\cos^4 x & 3+2\sin^4 x & \sin^2 2x \end{vmatrix}$, then $\frac{1}{5}f'(0)=$

1) 0 **2)** 1 **3)** 2 **4)** 6

13. Let $I_1=\int_0^1 \frac{e^x}{1+x}dx$ and $I_2=\int_0^1 \frac{x^2}{e^{x^3}\left(2-x^3\right)}dx$, then $\frac{I_1}{I_2}=$

1) $\frac{3}{e}$ **2)** $\frac{1}{3e}$ **3)** $\frac{e}{3}$ **4)** 3e

14. If z_1 and z_2 both satisfy the equations $\mathrm{Re}(z)=|z-2|$ and $Arg\left(z_1-z_2\right)=\frac{\pi}{3}$ then $I_m\left(z_1+z_2\right)$ is

1) $\frac{1}{\sqrt{3}}$ **2)** $\frac{2}{\sqrt{3}}$ **3)** $\sqrt{3}$ **4)** $\frac{4}{\sqrt{3}}$

15. A circle touches y-axis at (0,3) and makes an intercept of 2 units on the positive x-axis. Then intercept made by the circle on the line $\sqrt{10}x-3y=1$ is

1) 3 **2)** 6 **3)** $2\sqrt{10}$ **4)** $\sqrt{10}$

16. Match the eccentricity corresponding ellipse

Column-I	Column-II
i) $4x^2+9y^2-24x+36y-72=0$	P) $\frac{\sqrt{5}}{3}$
ii) $9x^2+25y^2-18x-10y-116=0$	Q) $\frac{\sqrt{3}}{2}$
iii) $36x^2+144y^2-36x-96y-119=0$	R) $\frac{4}{5}$
iv) $9x^2+5y^2-30y=0$	S) $\frac{2}{3}$

1) i-P,ii-R,iii-Q,iv-S **2)** i-P,ii-R,iii-S,iv-Q

3) i-P,ii-Q,iii-R,iv-S **4)** i-P,ii-Q,iii-S,iv-R

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17. Let R be the set of real number consider the following subsets of the plane $R\times R$

Statement -I : S = $\{(x,y)/\ y=x+1, 0<x<2\}$ is an equivalence relation

Statement -II: T = $\{(x,y)/\ x-y$ is an integer$\}$ is an equivalence relation

1) Both statement -I and statement -II are true

2) Statement -I is true, statement -II is false

3) Statement -I is false, statement -II is true

4) Both statement -I, statement II are false

18. If $I=\int\frac{2x^2}{4+x^2}dx=2x-f(x)+c$ where $f(2)=\pi$, then the minimum value of $y=f(x)\forall x\in[-2,2]$ is

1) 0 **2)** $-\pi$ **3)** 2π **4)** -4π

19. The area bounded by $y=\min\{x,2-x\}$ and $y=(x-1)^2$ is K then $[K]$ is equal to (where [.] is greatest integer function)

1) 1 **2)** 3 **3)** 0 **4)** 2

20. Let $y=y(x)$ solution of differential equation

$$\frac{dy}{dx}+\frac{\sqrt{2}y}{2\cos^4 x-\cos 2x}=x.e^{\text{Tan}^{-1}}\left(\sqrt{2}\cot 2x\right),0<x<\frac{\pi}{2}$$

with $y\left(\frac{\pi}{4}\right)=\frac{\pi^2}{32}$. If $y\left(\frac{\pi}{3}\right)=\frac{\pi^2}{18}e^{-\text{Tan}^{-1}(\alpha)}$, then $3\alpha^2=$

1) 1 **2)** 2 **3)** 3 **4)** 4

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

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21. The least value of n such that $(n-2)x^2+8x+n+4>0\ \forall x\in R,\ n\in N$ is

22. If the function $f(x)=\begin{cases}\dfrac{1}{|x|} & |x|\geq 2\\ ax^2+2b & |x|<2\end{cases}$ is differentiable on R then $48(a+b)$ is

23. If the latusrectum of the hyperbola $\dfrac{x^2}{9}-\dfrac{y^2}{b^2}=1$ subtend an angle of $\dfrac{\pi}{3}$ at the centre of the hyperbola If b^2 is equal to $\dfrac{\ell}{m}\left(1+\sqrt{n}\right)$ where ℓ, m are co-prime numbers then $\ell^2+m^2+n^2$ is equal to

24. 8 cards and 8 envelopes are numbered 1,2,3,4,5,6,7,8 and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same numbers and moreover the card numbered 2 and 4 always placed in the envelopes numbered 1 and 3 respectively, then number of ways of placing cards in envelopes with above constraint is

25. A die is tossed If the die shows 1 or 2 then a fair coin is tossed and if the die slows 3 then two fair coins are tossed other wise the three fair coins are tossed given that the resulting fair coin tossed produced no heads. If the probability that the die showed 1 or 2 is $\dfrac{m}{n}$ (where m, n are co-prime) then the value of $m+n$ is

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

26. Match the following

List –I	**List –II**
A) Moderator	e) Absorbs heat
B) Control rods	f) Prevents radiation from getting exposed to the outside
C) Radiation shielding	g) Absorb neutrons
D) Coolant	h) Slow down neutrons

1) A-h, B-g, C-f, D-e **2)** A-g, B-h, C-f, D-e

3) A-h, B-g, C-e, D-f **4)** A-h, B-f, C-e, D-g

27. Assertion : A current I flows along the length of an infinitely straight and thin walled pipe. Then the magnetic field at any point inside the pipe is zero

Reason : $\oint \vec{B}.d\vec{l} = \mu_0 I$

Read the assertion and reason carefully to mark the correct option out of the options given below

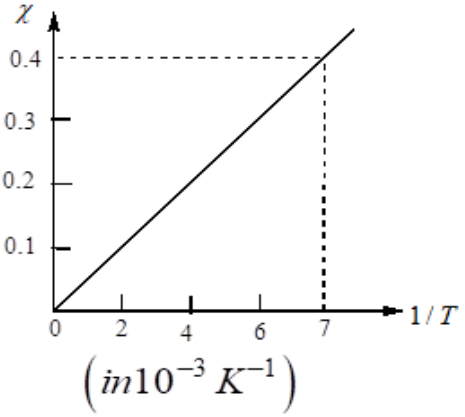
1) Assertion and Reason both are false

2) Both Assertion and Reason are true but reason is not the correct explanation of the assertion

3) Assertion is true but Reason is false

4) Both Assertion and Reason are true and the reason is the correct explanation of the Assertion

28. The χ-1/T graph for an alloy of paramagnetic nature is shown in figure. The curie constant is nearly



1) $2.8\times10^{-3}K$ **2)** 57 K **3)** 570 k **4)** $17.5\times10^{-3}k$

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29. **Statement 1:** A large soap bubble expands while a small bubble shrinks, when they are conceted to each other by a capillary tube

Statement 2: The excess pressure inside bubble is inversely proportional to its radius

1) Statement -1 is True, Statement -2 is True; Statement -2 is a correct explanation of Statement -1

2) Statement -1 is True, Statement-2 is True; Statement -2 is not a correct explanation for statement -1

3) Statement -1 is True, Statement -2 is False

4) Statement -1 is False, Statement -2 is True

30. Magnetic field associated with electromagnetic wave whose electric filed is given by $\vec{E} = 2.1\sin\left(3\times10^8 t - 1.8Z\right)\hat{i}N/C$, is

1) $\vec{B} = 1.26\times10^{-8}\sin\left(3\times10^8 t - 1.8Z\right)\hat{j}T$

2) $\vec{B} = 1.8\times10^{-8}\sin\left(3\times10^8 t - 1.8Z\right)\hat{j}T$

3) $\vec{B} = 0.7\times10^{-8}\sin\left(3\times10^8 t - 1.8Z\right)\hat{j}T$

4) $\vec{B} = -1.26\times10^{-8}\sin\left(3\times10^8 t - 1.8Z\right)\hat{j}T$

31. In a standard YDSE the distance between the slits varies with time $(t)as\ d(t) = d_0 + a_0\sin(\omega t)$. The difference between the largest fringe width and the smallest fringe width over time is given as (λ is wavelength of monochromatic light used, D is the screen distance from slits, D>>d)

1) $\frac{2\lambda D d_0}{d_0^2 - a_0^2}$ **2)** $\frac{2\lambda D a_0}{d_0^2 - a_0^2}$ **3)** $\frac{\lambda D}{d_0 + a_0}$ **4)** $\frac{\lambda D a_0}{d_0^2 - a_0^2}$

32. A particle of mass m is moving in the xy-plane such that its velocity at a point (x, y) is given as $\vec{v} = \alpha\left(y\hat{x} + 2x\hat{y}\right)$, where α is a non-zero constant. What is the force $\vec{F}$ acting on the particle?

1) $\vec{F} = 2m\alpha^2\left(x\hat{x} + y\hat{y}\right)$ **2)** $\vec{F} = m\alpha^2\left(y\hat{x} + 2x\hat{y}\right)$

3) $\vec{F} = 2m\alpha^2\left(y\hat{x} + x\hat{y}\right)$ **4)** $\vec{F} = m\alpha^2\left(x\hat{x} + 2y\hat{y}\right)$

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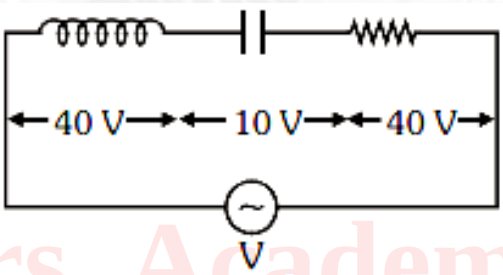
33. In an organ pipe successive resonance are obtained at 250 Hz, 350 Hz and 450 Hz. If the speed of sound is 300 m/s, then length of organ pipe (in m) is x, then the value of 10x is ____ (ignore end correction)

1) 10 **2)** 25 **3)** 5 **4)** 15

34. A parallel plate capacitor had a uniform electric '$\overline{E}$' in the space between the plates. If the distance between the plates is 'd' and area of each plate is 'A'. The energy density in the capacitors is: (ε_0 =permittivity of free space)

1) $\varepsilon_0 EAd$ **2)** $\frac{1}{2}\varepsilon_0 E^2$ **3)** $\frac{1}{2}\varepsilon_0 E^2 Ad$ **4)** $\frac{E^2 Ad}{\varepsilon_0}$

35. An inductor of inductance L, a capacitor of capacitance C and a resistor of resistance 'R' are connected in series to an ac source of potential difference 'V' volts as shown in figure. Potential difference across L, C and R is 40 V, 10 V and 40 V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}$A. the impedance of the circuit in Ohm is



1) 0 **2)** 50 **3)** 5 **4)** 10

36. An α - particle and proton are accelerated from rest by a potential difference of 100V. After this their de-Broglie wave lengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$ is

1) $\frac{2\sqrt{2}}{1}$ **2)** $\frac{1}{2\sqrt{2}}$ **3)** $\frac{\sqrt{2}}{1}$ **4)** $\frac{1}{\sqrt{2}}$

37. A flat horizontal board moves up and down in SHM of amplitude A. Then the shortest permissible time period of the vibration such that an object placed on the board may not lose contact with the board is

1) $2\pi\sqrt{\frac{g}{A}}$ **2)** $2\pi\sqrt{\frac{A}{g}}$ **3)** $2\pi\sqrt{g\times A}$ **4)** $\frac{1}{2\pi}\sqrt{\frac{g}{A}}$

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38. A liquid is kept in a cylindrical vessel which is rotated along its axis. The liquid rises at the sides. If the radius of the vessel is 0.05m and the speed of rotation is 2 rev/sec. Find the difference in the height of the liquid at the centre of the vessel and its sides (in cm)

Take g = 10 m /s^2 & $\pi^2 = 10$

1) 1 **2)** 2 **3)** 3 **4)** 4

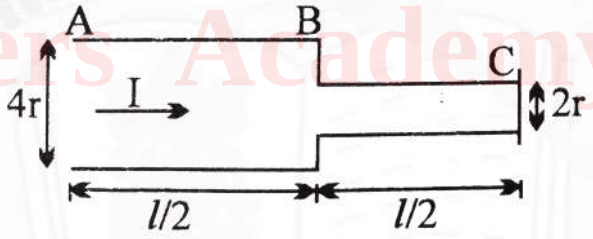
39. Two identical glass $\left(\mu_g = \frac{3}{2}\right)$ equi-convex lenses of focal length f are kept in contact.

The space between the two lenses is filled with water $\left(\mu_w = \frac{4}{3}\right)$. The focal length of the combination

1) f **2)** $\frac{f}{2}$ **3)** $\frac{4f}{3}$ **4)** $\frac{3f}{4}$

40. Consider a cylindrical element as shown in the adjoining figure. Current flowing through the element is I and resistivity of the material of the cylinder is ρ.

Choose the correct option out of the following.



1) Drift velocity in the second half is equal to that of in the first half

2) Current density in both halves are equal

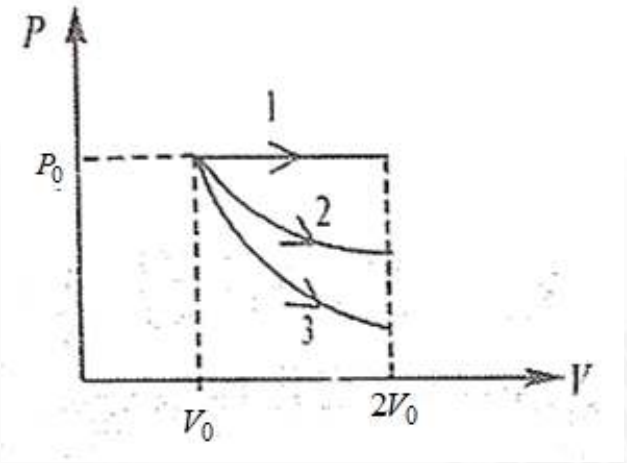
3) Electric field in both halves is equal.

4) Voltage drop in first half is ¼ of voltage drop in second half.

41. A gas is expanded from volume V_0 to $2V_0$ under three different processes. Process 1 is isobaric process, process 2 is isothermal and process 3 is adiabatic. Let $\Delta U_1, \Delta U_2$ and ΔU_3 be the change in internal energy of the gas in these three processes. Then

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1) $\Delta U_1 > \Delta U_2 > \Delta U_3$ **2)** $\Delta U_1 < \Delta U_2 < \Delta U_3$ **3)** $\Delta U_2 < \Delta U_1 < \Delta U_3$ **4)** $\Delta U_2 < \Delta U_3 < \Delta U_1$

42. A chain of length L and mass M is arranged as shown in following five cases. In fig iv chain covers half of the semi-circle. The correct decreasing order of potential energy is (Assumed zero at horizontal surface)



1) $i > ii > iii > iv > v$ **2)** $i = ii > iii > iv > v$

3) $i = ii > iv > iii > v$ **4)** $i = ii > iii > v > iv$

43. In Searle's apparatus, used for study of extension of wires verses load, we have two wires. During the experiment, we study the extension in one wire. The use of second wire is _____

1) To support the device/ apparatus because, it is heavy and may break the single wire.

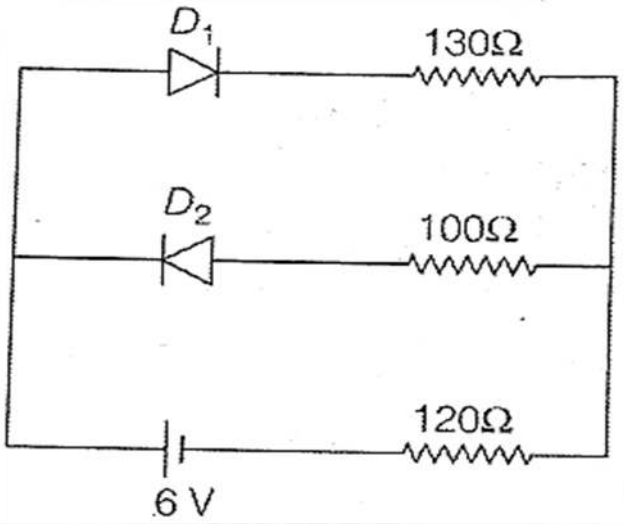
2) To compensate for the change in length caused, by changes in temperature of atmosphere during experimentation.

3) To keep the apparatus in level so that extension is measured accurately.

4) To prevent unnecessary twisting of the instrument during experimentation.

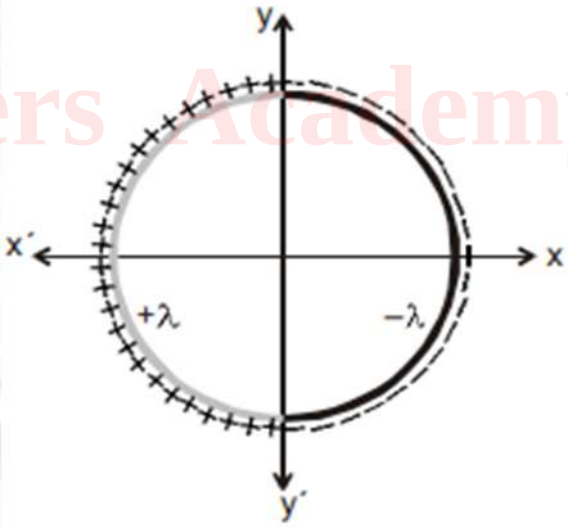
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44. The circuit contains two diodes each with a forward resistance of 50Ω and with infinite reverse resistance. If the battery voltage is 6V, the current through the 120Ω resistance is.....mA



1) 0 **2)** 5 **3)** 15 **4)** 20

45. A thin ring of radius R metres is placed in x-y plane such that its centre lies at the origin. The half ring in region x<0 carries uniform linear charge density $+\lambda C/m$ and the remaining half ring in region x>0 carries uniform linear charge density $-\lambda C/m$.



The electric potential (in volts) at point P whose coordinates are $\left(0m,+\frac{R}{2}m\right)$ is

1) $\frac{1}{4\pi\varepsilon_0}\frac{\lambda}{4}$ **2)** $\frac{1}{4\pi\varepsilon_0}\frac{\lambda}{2}$ **3)** Zero **4)** $\frac{1}{4\pi\varepsilon_0}\lambda$

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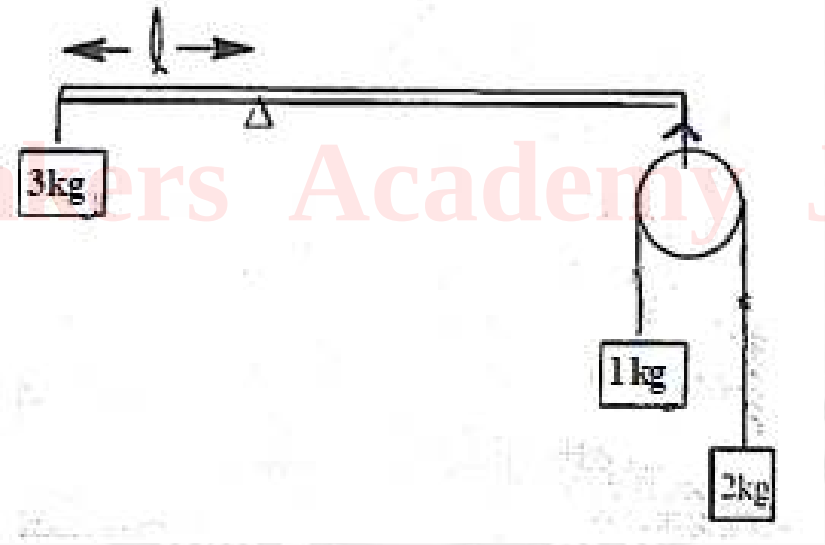
SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

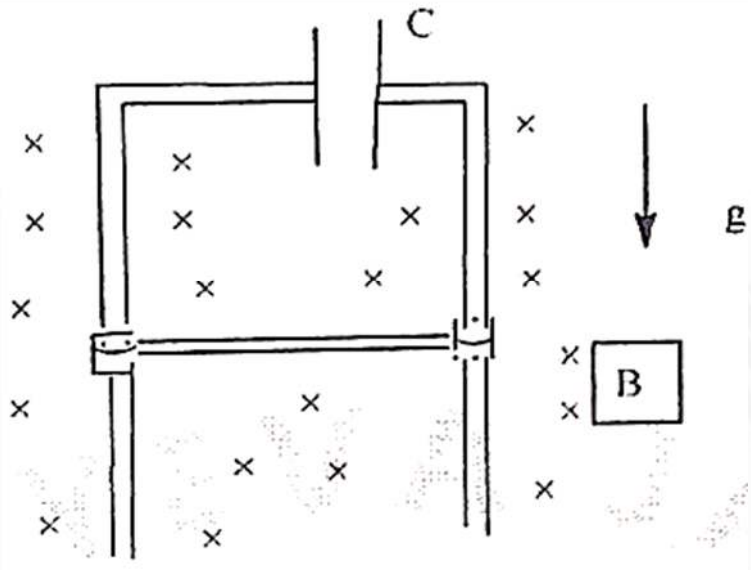
46. If mass density of earth varies with distance 'r' from centre of earth as $\rho = kr$ and 'R' is radius of earth. The orbital velocity of an object revolving around earth at a distance '2R' from its centre is $\sqrt{\frac{\pi k R^x G}{y}}$, then $x + y =$ _____

47. A meter scale of negligible mass is pivoted at a distance (ℓ) from the left end as shown, such that it remains horizontal. Take $g = 10 m/s^2$ and pulley to be mass less and frictionless. Threads are massless. If $\ell = \left(\frac{x}{y}\right)$ meters, then (y-x) is _____ (where x and y are smallest possible integers)



48. A capacitor (C = 4 farad) is connected, as shown to a vertical conducting frame, where a horizontal rod glides down without friction in uniform horizontal magnetic field. Take B =1 tesla; ℓ = 0.5 meter, mass of rod (m) = 1 kg and $g = 10 m/s^2$. The acceleration of the rod while gliding down is _____ m/s^2

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49. The mass of a hydrogen molecule is 3.23 x 10^{-27} kg. If 10^{23} hydrogen molecules strike on 2cm^2 area of a wall per second at an angle 45^0 with normal to the wall with a speed 10^5cms^{-1}, the pressure they exert on the wall is _____$\times 10^2\, Pa$. (Take $\sqrt{2}=1.4$)

50. A Screw gauge of pitch 0.5 mm is used to measure the diameter of uniform wire of length 6.9cm. The main scale reading is 1.5mm and circular scale reading is 7. The curved surface area of wire is $x\times 10^{-5}\, m^2$. Find 'x' [Screw gauge has to 50 div on its circular sale] (take $\pi=3.14$ & $\pi^2=10$)

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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

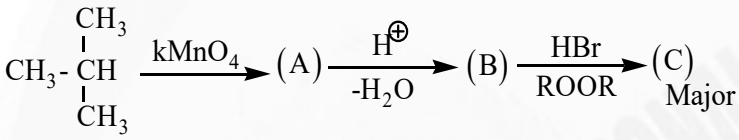
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. Lassaigne's test for the detection of nitrogen fails in

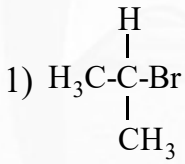
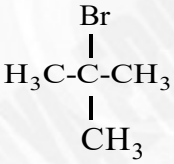
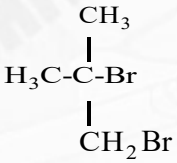
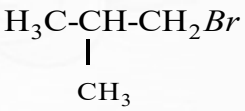
1) $NH_2\text{-}CO\text{-}NH\text{-}NH_2.HCl$ **2)** $H_2N\text{-}NH_2.HCl$

3) $H_2N\text{-}CO\text{-}NH_2$ **4)** $C_6H_5NH\text{-}NH_2.HCl$

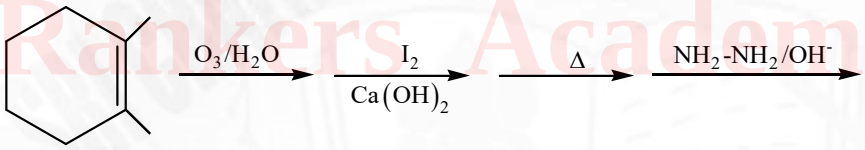
52.

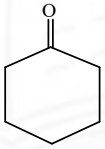
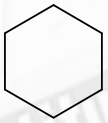


The product "c" in the above reaction formed is

1)  2)  3)  4) 

53. End product in the following sequence of reaction is



1)  2)  3)  4) 

54. Given below are two statements, one is labelled as **Assertion (A)** and other is labelled as **Reason (R) :**

Assertion (A) : Gabriel phthalimide synthesis can not be used to prepare aromatic primary amines

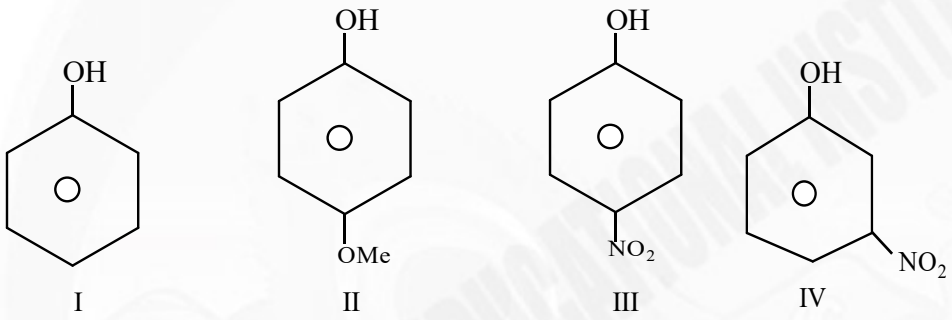
Reason (R) : Aryl halides do not undergo nucleophilic substitution reaction at room temperature

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In the light of the above statements, choose the correct answer from the option given below

1) Both (A) and (R) true but (R) is not the correct explanation of A
2) (A) is false but (R) is true
3) Both (A) and (R) true and (R) is correct explanation of A
4) (A) is true but (R) is false

55. The K_a of

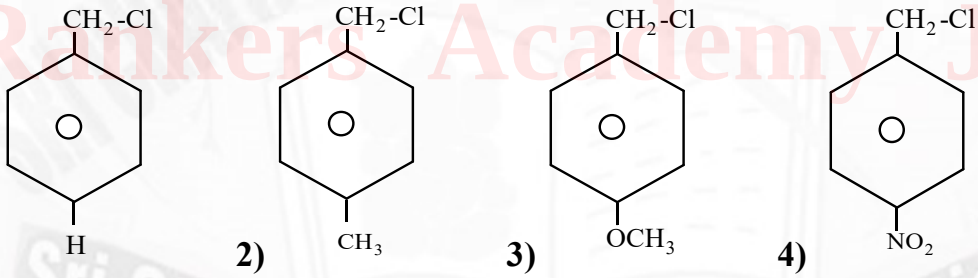


I II III IV

1) III > IV >I > II **2)** IV > III >II > I

3) II > I >IV > III **4)** III > II >IV > I

56. Which of the following is most reactive toward SN^2?



1) **2)** **3)** **4)**

57. Which of the following statement is not correct?

1) Number of H-bonds between Adenine and thymine are 2

2) α -D Glucopyranose and β-D Glucopyranose are anomers

3) Sucrose is non reducing sugar

4) All amino acids except Glycine are optically inactive

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58. Number of bridge ligands and coordination number in complex compound $\left[Co_2(CO)_8\right]$ is

1) 2, 6 **2)** 3, 6 **3)** 2, 4 **4)** 6, 3

59. Consider the following complex ions $P=\left[Fe\ F_6\right]^{3-}$ $Q=\left[V\left(H_2O\right)_6\right]^{2+}$ $R=\left[Fe\left(H_2O\right)_6\right]^{2+}$

The correct order of the complex ions , according to their spin only magnetic moment values (in BM) is :

1) Q < R< P **2)** R< Q <P **3)** R < P < Q **4)** Q < P < R

60. Which of the following statement is correct

1) In the formation of dioxygen from oxygen atoms, 10 molecular orbitals will be formed

2) All molecular orbitals in the dioxygen will be completely filled

3) Total number of bonding molecular orbitals will not be same as total number of anti bonding orbitals in dioxygen

4) Number of filled bonding orbitals will be same as number of filled anti bonding orbitals

61. Which of the following statement is incorrect

1) Highest oxidation state of manganese in fluorides is +4 (MnF_4) but highest oxidation state in oxides is +7 (Mn_2O_7)

2) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$ is the correct order of oxidizing power

3) When I^- is oxidized by MnO_4^- in alkaline medium, I^- converts in to I_2

4) Silver nitrate on heating decomposes to give two types of paramagnetic gases along with a residue

62. Given below are two statements

Statements I: $N(CH_3)_3$ and $P(CH_3)_3$ can act as ligands to from transition metal complexes

Statements II: As , N and P are from same group the nature of bonding of $N(CH_3)_3$ and $P(CH_3)_3$ is always same with transition metals

In the light of the above statements, choose the most appropriate answer from the options given below

1) Both statement I and statement II are incorrect

2) Statement I is incorrect but statement II is correct

3) Both statement I and statement II are correct

4) Statement I is correct but statement II is incorrect

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63. Match the column I with column II and mark the appropriate choice

	Column-I		Column-II
A)	Clemmensen reduction	i)	Conc. KOH
B)	Rosenmund reduction	ii)	Zn/Hg + Conc.HCl
C)	Iodoform reaction	iii)	$H_2/Pd\text{-}BaSO_4$
D)	Cannizzaro reaction	iv)	$NaOH+I_2$

1) A-i,B-iii,C-ii,D-iv **2)** A-iii,B-iv,C-i,D-ii

3) A-ii,B-iii,C-iv,D-i **4)** A-iv,B-i,C-ii,D-iii

64. For a first order reaction A→B time taken for 10% completion of reaction is 20 min. The time taken for 19% completion of the reaction is ___min

(given $\log^{10}=1; \log^{9}=0.954$)

1) 20 min **2)** 40 min **3)** 60 min **4)** 120 min

65. A gaseous mixture contains hydrogen atoms in the 4th excited state, He^{+} ions in 3rd excited state and Li^{+2} ions in 2nd excited state the number of spectral lines obtained in the emission spectrum of this sample when all these atoms/ions return to the ground state is

1) 19 **2)** 20 **3)** 16 **4)** 18

66. The precipitate of $CaF_2\left(k_{sp}=1.7\times10^{-10}\right)$ is obtained when equal volumes of each of the following is mixed at constant temperature

1) $1\times10^{-2}MCa^{+2}$ and $1\times10^{-3}M\ F^{-}$ **2)** $1\times10^{-4}MCa^{+2}$ and $1\times10^{-4}M\ F^{-}$

3) $1\times10^{-2}MCa^{+2}$ and $1\times10^{-5}M\ F^{-}$ **4)** $1\times10^{-3}MCa^{+2}$ and $1\times10^{-5}M\ F^{-}$

67. If the standard electrode potentials of Fe^{3+}/Fe and Fe^{+2}/Fe are -0.04V and -0.44V respectively then that of Fe^{+3}/Fe^{+2} is

1) 0.76 V **2)** -0.76 V **3)** 0.40 V **4)** -0.40 V

68. Among the following halogens F_2,Cl_2, Br_2 and I_2 which can undergo disproportion reactions

1) Cl_2, Br_2, and I_2 **2)** F_2 and Cl_2 **3)** F_2,Cl_2 and Br_2 **4)** only I_2

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69. For the given hypothetical reactions the equilibrium constants are as follows

$x \rightleftharpoons y; k_1 = 1.0,$

$y \rightleftharpoons z; k_2 = 2.0,,$

$z \rightleftharpoons w; k_3 = 4.0,$

The equilibrium constant for the reaction

$x \rightleftharpoons w$ is

1) 12.0 **2)** 8.0 **3)** 7.0 **4)** 6.0

70. The use of H_2S as regent for qualitative analysis of cations is based on the fact that

1) H_2S forms colored sulphides

2) H_2S is a weak acid precipitating agent and we can change degree of ionization of H_2S by adding HCl and NH_4OH

3) H_2S is a weak acid

4) The colour of metal sulphide precipitated depends upon the P^H of the solution

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. An amine (x) is prepared by ammonolysis of benzyl chloride on adding P-toluene sulphonyl chloride to it the solution remains clear Molar mass of the amine (x) formed is ____$gmol^{-1}$

(Given molar mas in g mol^{-1} C:12, H:1, O:16, N:14)

72. A solution containing 10 g of an electrolyte AB_2 in 100 g of water boils at 100.52°C. The degree of ionization of electrolyte (α) is _____$\times 10^{-1}$

(Given : molar mass of AB_2 200 g mol^{-1} K_b of water = 0.52 K kg mol^{-1} B.P of water 100°c ; $AB_{2(S)}$ ionises as $AB_{2(s)} \rightleftharpoons A^{+2}_{(aq)} + 2B^{-}_{(aq)}$

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73. 3.92 g of Mohr's salt react completely with 50 ml $\frac{N}{10}$ $KMnO_4$ solution in acidic medium the percentage purity of the sample is x% then x is

(Given atomic weight of Fe = 56g, N =14 g, H=1g, S=32g, O = 16g)

74. The heat of dissociation of $H_{2(g)}$ is 435 KJmol^{-1} If $C_{(graphite)} \rightarrow C(gas)\Delta H = 720\ kJ\ mol^{-1} and\ \Delta_f^o H$ For CH_4=-75 kJ mol^{-1} then $\Delta_a^o H$ of $CH_{4(g)}$ is 'x' kJ mol^{-1} then the value of $\frac{x}{4}$ is

75. The total no. of species among the following which have H-bonding – $NH_{3(g)}$, $H_2O_{(g)}, HF_{(g)}, H_2O_{2(l)}, CH_3COOH_{(l)}, NaHCO_3, KHCO_3$, O-fluorophenol, O-nitrophenol, P-nitrophenol, salicylaldehyde, chloral hydrate, salicylic acid, glycol, glycerol

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